Chapter 1. Linear Regression and Gradient Descent

Regression, normal equations, gradient descent, generalisation

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1 The Regression Problem

In regression we observe pairs $(\mathbf{x}_i, y_i)$, $i = 1, \dots, n$, with input $\mathbf{x}_i \in \mathbb{R}^k$ and real response y_i, and we want a function that predicts y from $\mathbf{x}$. The simplest choice is a linear function. We motivate the choice of "best line" by the sum of squared errors (SSE): a horizontal benchmark line gives one SSE, rotating it changes the SSE, and plotting SSE against the rotation angle gives a curve that first decreases and then increases. The best line is the minimiser of this curve.

Definition 1.1: linear regression model and design matrix

$$y_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} + \varepsilon_i, \qquad i = 1, \dots, n.$$

Absorb the intercept with a constant input 1: let $d = k + 1$, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)^\top \in \mathbb{R}^d$, and let the **design matrix** $X \in \mathbb{R}^{n \times d}$ have ith row $(1, x_{i1}, \dots, x_{ik})$. Then

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad f(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2 = (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

Rows of X index samples; columns index features.

2 Matrix Calculus Toolkit

We use the gradient convention: the gradient of a scalar with respect to a vector is a column vector of the same size.

Theorem 1.2: basic gradients

For $\mathbf{a} \in \mathbb{R}^d$ and $A \in \mathbb{R}^{d \times d}$,

$$\nabla_{\boldsymbol{\beta}} (\mathbf{a}^\top \boldsymbol{\beta}) = \mathbf{a}, \qquad \nabla_{\boldsymbol{\beta}} (\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = (A + A^\top)\boldsymbol{\beta}, \qquad \nabla^2_{\boldsymbol{\beta}} (\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = A + A^\top.$$

If A is symmetric the last two become $2A\boldsymbol{\beta}$ and $2A$.

Proof. $\mathbf{a}^\top \boldsymbol{\beta} = \sum_j a_j \beta_j$, so $\partial / \partial \beta_k = a_k$. For the quadratic form, $\boldsymbol{\beta}^\top A \boldsymbol{\beta} = \sum_{i,j} A_{ij} \beta_i \beta_j$. Differentiating with respect to β_k, the product rule picks up the terms with $i = k$ and those with $j = k$:

$$\frac{\partial}{\partial \beta_k} \sum_{i,j} A_{ij} \beta_i \beta_j = \sum_j A_{kj} \beta_j + \sum_i A_{ik} \beta_i = (A\boldsymbol{\beta})_k + (A^\top \boldsymbol{\beta})_k.$$

Differentiating once more gives the constant matrix $A + A^\top$. ■

3 Normal Equations

Theorem 1.3: least squares solution

$\nabla f(\boldsymbol{\beta}) = 2X^\top (X\boldsymbol{\beta} - \mathbf{y})$ and $\nabla^2 f = 2X^\top X \succeq 0$. Every minimiser solves the **normal equations**

$$X^\top X \boldsymbol{\beta} = X^\top \mathbf{y}.$$

If X has full column rank d, then $X^\top X \succ 0$ and the unique minimiser is $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$.

Proof. Expand $f = \mathbf{y}^\top\mathbf{y} - 2\mathbf{y}^\top X\boldsymbol{\beta} + \boldsymbol{\beta}^\top X^\top X\boldsymbol{\beta}$ (the two cross terms are equal scalars). With the toolkit, $\nabla f = -2X^\top\mathbf{y} + 2X^\top X\boldsymbol{\beta}$ and $\nabla^2 f = 2X^\top X$. For every $\mathbf{v}$, $\mathbf{v}^\top X^\top X\mathbf{v} = \|X\mathbf{v}\|^2 \geq 0$, so f is convex and stationary points are global minimisers. If the columns of X are linearly independent, $X\mathbf{v} \neq \mathbf{0}$ whenever $\mathbf{v} \neq \mathbf{0}$; then $X^\top X \succ 0$, it is invertible, f is strictly convex and the minimiser is unique. ■

Remark 1.4: geometry and rank

The normal equations say $X^\top(\mathbf{y} - X\hat{\boldsymbol{\beta}}) = \mathbf{0}$: the residual is orthogonal to every column of X, so $X\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space. Since the first column is $\mathbf{1}_n$, the residuals sum to zero. Having $n > d$ does not by itself guarantee full rank; with dependent columns the minimiser is not unique and the pseudoinverse gives the minimum-norm solution. Numerically one solves the linear system by QR or SVD rather than forming an inverse.

Example 1.5: least squares on three points

Points $(0,1),(1,2),(2,2)$ and model $y = \beta_0 + \beta_1 x$:

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad X^\top X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad X^\top\mathbf{y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad (X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}.$$

Hence $\hat{\boldsymbol{\beta}} = \frac{1}{6}(25-18, -15+18)^\top = (7/6,\, 1/2)^\top$. Predictions are $(7/6, 5/3, 13/6)$, residuals $(-1/6, 1/3, -1/6)$ (sum zero), and SSE $= 1/6$.

4 Gradient Descent

4.1 The algorithm

Instead of solving the normal equations directly we may iterate

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta\nabla f(\boldsymbol{\beta}_t) = \boldsymbol{\beta}_t + 2\eta X^\top(\mathbf{y} - X\boldsymbol{\beta}_t),$$

i.e. componentwise $\beta_j \leftarrow \beta_j + 2\eta\sum_i (y_i - \hat{y}_i)x_{ij}$ with $x_{i0} = 1$. The factor 2 is often absorbed into the learning rate η. Each full-batch step uses all samples. The negative gradient is the direction of steepest descent: for a unit vector $\mathbf{v}$ the first-order change $\nabla f^\top\mathbf{v}$ is minimised, by Cauchy–Schwarz, at $\mathbf{v} = -\nabla f/\|\nabla f\|$.

4.2 Convergence on least squares

Write $J(\boldsymbol{\beta}) = \frac{1}{2}\|X\boldsymbol{\beta} - \mathbf{y}\|^2$ (so $\nabla J = X^\top(X\boldsymbol{\beta} - \mathbf{y})$) and $H = X^\top X$, assumed positive definite.

Theorem 1.6: exact convergence condition

Gradient descent $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta\nabla J(\boldsymbol{\beta}_t)$ converges to $\hat{\boldsymbol{\beta}}$ for every initial point if and only if

$$0 < \eta < \frac{2}{\lambda_{\max}(X^\top X)}.$$

Proof. Error recursion. Since $\nabla J(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, $X^\top\mathbf{y} = H\hat{\boldsymbol{\beta}}$, so $\nabla J(\boldsymbol{\beta}_t) = H(\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}})$. With $\mathbf{e}_t = \boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}$,

$$\mathbf{e}_{t+1} = \mathbf{e}_t - \eta H\mathbf{e}_t = (\mathbf{I} - \eta H)\mathbf{e}_t, \qquad \mathbf{e}_t = (\mathbf{I} - \eta H)^t\mathbf{e}_0.$$

Diagonalise. H is symmetric, so $H = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$, $\lambda_i > 0$. Using $\mathbf{I} = QQ^\top$, $\mathbf{I} - \eta H = Q(\mathbf{I} - \eta\Lambda)Q^\top$. Let $\mathbf{c}_t = Q^\top \mathbf{e}_t$. Multiplying the recursion by $Q^\top$ and using $Q^\top Q = \mathbf{I}$ gives $\mathbf{c}_{t+1} = (\mathbf{I} - \eta\Lambda)\mathbf{c}_t$, a diagonal recursion:

$$c_{t,i} = (1 - \eta\lambda_i)^t \, c_{0,i}.$$

Because $\|\mathbf{c}_t\| = \|\mathbf{e}_t\|$, $\mathbf{e}_t \to \mathbf{0}$ iff every coordinate $c_{t,i} \to 0$.

Sufficiency. If $|1 - \eta\lambda_i| < 1$ for all i, every coordinate tends to zero for every $\mathbf{e}_0$.

Necessity. If $|1-\eta\lambda_i| \geq 1$ for some i, start from $\mathbf{e}_0 = \mathbf{q}_i$ (the ith eigenvector). Then $\mathbf{e}_t = (1-\eta\lambda_i)^t \mathbf{q}_i$ and $\|\mathbf{e}_t\| = |1 - \eta\lambda_i|^t \geq 1$, so this initial point does not converge.

Solve. $|1 - \eta\lambda_i| < 1 \iff 0 < \eta\lambda_i < 2 \iff 0 < \eta < 2/\lambda_i$. Requiring this for all i is equivalent to $0 < \eta < 2/\lambda_{\max}$. ■

Remark 1.7: conditioning

The contraction factor of direction i is $|1 - \eta\lambda_i|$. The rate $\max_i |1 - \eta\lambda_i|$ is minimised by $\eta^\star = 2/(\lambda_{\max} + \lambda_{\min})$, giving rate $(\kappa - 1)/(\kappa + 1)$ with condition number $\kappa = \lambda_{\max}/\lambda_{\min}$. Elongated (ill-conditioned) level sets make gradient descent slow; this is revisited in Chapter 8. At $\eta = 2/\lambda_{\max}$ the steepest direction oscillates with factor -1 and never decays, so the inequality is strict. In Example 1.5, $\lambda(X^\top X) = 4 \pm \sqrt{10}$ and the condition is $0 < \eta < (4 - \sqrt{10})/3 \approx 0.279$.

5 Polynomial Features and Generalization

5.1 Feature maps

For scalar input take $\boldsymbol{\phi}_M(x) = (1, x, x^2, \dots, x^M)^\top$ and $y(x) = \sum_{j=0}^{M} \beta_j x^j$. The model is nonlinear in x but linear in the coefficients, so all of the above applies with the (Vandermonde-type) design matrix whose rows are $\boldsymbol{\phi}_M(x_i)^\top$.

5.2 Training error versus generalization

Because a degree-M model contains every lower-degree model, the minimal training SSE cannot increase as M increases. Validation error can nevertheless increase: a high-degree fit follows noise and oscillates between data points (overfitting), while too low a degree cannot capture the trend (underfitting). Performance is measured by the root mean squared error

$$\mathrm{MSE} = \frac{1}{n}\mathrm{SSE}, \qquad E_{\mathrm{RMS}} = \sqrt{\mathrm{MSE}}.$$

Remark 1.8: roles of the data splits

Training data choose the coefficients; validation data choose the degree and the regularisation strength; a held-out test set estimates final performance. In typical learning curves the gap between training and validation RMS grows with M for $M \geq 5$, and degree 9 with $n = 10$ overfits badly while $n = 100$ points support the same degree much better. "About ten points per parameter" is a heuristic, not a theorem. Regularisation (Chapter 4) is the second remedy.

6 Proof Supplement: Uniqueness and the Pseudoinverse

Theorem 1.9: set of minimisers

The minimisers of $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ are exactly the solutions of $X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}$; this system is always consistent, and the solution set is $\hat{\boldsymbol{\beta}}_0 + \ker X$ for any particular solution $\hat{\boldsymbol{\beta}}_0$. Among them, $X^+\mathbf{y}$ has minimal Euclidean norm.

Proof. Consistency: $X^\top \mathbf{y}$ lies in the column space of $X^\top$, which equals the column space of $X^\top X$ (both have orthogonal complement $\ker X$, since $X^\top X\mathbf{v} = \mathbf{0} \Rightarrow \|X\mathbf{v}\|^2 = 0$). Two solutions differ by $\mathbf{v}$ with $X^\top X\mathbf{v} = \mathbf{0}$, i.e. $\mathbf{v} \in \ker X$, and give the same fitted values. Write $X = U\Sigma V^\top$ (thin SVD with r positive singular values). Every solution is $V_r\Sigma_r^{-1}U_r^\top \mathbf{y} + \mathbf{v}$ with $\mathbf{v} \perp \text{range}(V_r)$; by Pythagoras its norm is minimised at $\mathbf{v} = \mathbf{0}$, which is $X^+\mathbf{y}$. ■

Chapter 2. Probability, Likelihood and Bayesian Estimation

Random variables, maximum likelihood, MAP estimation, Bayesian models

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1 Events and Conditional Probability

A sample space Ω contains the possible outcomes and events are (measurable) subsets. The axioms are nonnegativity, $P(\Omega) = 1$ and countable additivity for disjoint events. They imply

$$P(\emptyset) = 0, \qquad P(E^c) = 1 - P(E), \qquad P(E \cup F) = P(E) + P(F) - P(E \cap F),$$

monotonicity $E \subseteq F \Rightarrow P(E) \le P(F)$, and continuity along increasing or decreasing sequences of events.

Definition 2.1: conditional probability, Bayes' rule, independence

For $P(E) > 0$, $P(F \mid E) = P(F \cap E)/P(E)$. For a partition $E_1, E_2, \dots$ of Ω,

$$P(F) = \sum_i P(F \mid E_i)P(E_i), \qquad P(E_i \mid F) = \frac{P(F \mid E_i)P(E_i)}{\sum_j P(F \mid E_j)P(E_j)}.$$

E, F are independent if $P(E \cap F) = P(E)P(F)$.

Example 2.2: why the prior matters

Let $P(D) = 0.01$, $P(+ \mid D) = 0.9$, $P(+ \mid D^c) = 0.1$. Then

$$P(D \mid +) = \frac{0.9(0.01)}{0.9(0.01) + 0.1(0.99)} \approx 0.0833.$$

A sensitive test does not make the posterior large when the prior is small. Two fair coin tosses: "first head" and "second head" each have probability $1/2$ and their intersection $1/4$, so they are independent.

2 Random Variables and Distributions

A random variable is a measurable function $X : \Omega \to \mathbb{R}$ (e.g. the number of heads in three tosses, $X(HHT) = 2$). Its CDF is $F_X(t) = P(X \le t)$.

- Discrete: probability mass $p(x) = P(X = x)$, $F_X(t) = \sum_{x \le t} p(x)$.
- Continuous: density $f \ge 0$ with $\int f = 1$, $P(a < X \le b) = \int_a^b f(u)\,du$. A density value is not a probability and may exceed one; $P(X = x) = 0$.

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \text{ or } \int g(x)f(x)\,dx, \qquad \operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Theorem 2.3: tools used throughout the course

(i) Linearity: $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$ always.

(ii) Independence: $\mathbb{E}[XY] = \mathbb{E}X\,\mathbb{E}Y$; hence $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$. Uncorrelatedness ($\operatorname{Cov} = 0$) suffices for the variance identity.

(iii) Tower property: $\mathbb{E}[\mathbb{E}[Z \mid X]] = \mathbb{E}Z$.

(iv) $\mathbb{E}[(Z - c)^2] = \operatorname{Var}Z + (\mathbb{E}Z - c)^2 \ge \operatorname{Var}Z$ for every constant c.

Proof. (iv): write $Z - c = (Z - \mathbb{E}Z) + (\mathbb{E}Z - c)$, square, and use $\mathbb{E}[Z - \mathbb{E}Z] = 0$ to kill the cross term. The other items are standard. ∎

3 Maximum Likelihood Estimation

Definition 2.4: likelihood and MLE

For iid data $x_1, \dots, x_n$ from $p(x \mid \theta)$,

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta), \qquad \hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \sum_{i=1}^{n} \log p(x_i \mid \theta).$$

The data are fixed and the parameter varies; L is not a distribution over θ. The logarithm preserves the maximiser and turns products into sums.

Example 2.5: Bernoulli, including the boundary

$x_i \in \{0, 1\}$ with $P(x_i = 1) = \theta$; let $s = \sum x_i$, $f = n - s$. Then $\log L = s \log \theta + f \log(1 - \theta)$. For $0 < s < n$,

$$\frac{d}{d\theta} \log L = \frac{s}{\theta} - \frac{f}{1 - \theta} = 0 \iff \theta = \frac{s}{n}, \qquad \frac{d^2}{d\theta^2} \log L < 0.$$

If $s = 0$, $L = (1 - \theta)^n$ is decreasing and the maximiser is 0; if $s = n$ it is 1. Both agree with s/n, but the derivative argument cannot be used there. Eight heads in ten tosses gives 0.8; two positive reviews out of two gives the extreme estimate 1.

Example 2.6: Gaussian

For $x_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$, giving $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ (biased; the unbiased version divides by $n - 1$).

4 Bayesian Inference and MAP

Definition 2.7: prior, posterior, evidence, MAP

With prior $p(\theta)$ and likelihood $p(D \mid \theta)$,

$$p(\theta \mid D) = \frac{p(D \mid \theta) p(\theta)}{p(D)}, \qquad p(D) = \int p(D \mid \theta) p(\theta) \, d\theta,$$

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \{\log p(D \mid \theta) + \log p(\theta)\}.$$

The evidence does not depend on θ. MAP is the posterior mode; the posterior mean is a different summary.

4.1 Beta–Bernoulli conjugacy

Theorem 2.8: Beta prior

If $\theta \sim \mathrm{Beta}(a, b)$, i.e. $p(\theta) = \theta^{a-1}(1 - \theta)^{b-1} / B(a, b)$ with $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$, and s

successes, f failures are observed, then

$$\theta \mid D \sim \mathrm{Beta}(a+s,\, b+f), \qquad \mathbb{E}[\theta \mid D] = \frac{a+s}{a+b+n}, \qquad \hat{\theta}_{\mathrm{MAP}} = \frac{a+s-1}{a+b+n-2}$$

(the mode formula needs both posterior parameters above one).

Proof. $p(\theta \mid D) \propto \theta^s(1-\theta)^f \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+s)-1}(1-\theta)^{(b+f)-1}$, which is the $\mathrm{Beta}(a+s, b+f)$ kernel. For $\mathrm{Beta}(\alpha,\beta)$, $\mathbb{E}\theta = B(\alpha+1,\beta)/B(\alpha,\beta) = \alpha/(\alpha+\beta)$ using $\Gamma(z+1) = z\Gamma(z)$. The mode solves $\frac{\alpha-1}{\theta} = \frac{\beta-1}{1-\theta}$, i.e. $\theta = \frac{\alpha-1}{\alpha+\beta-2}$. ■

The uniform prior is $\mathrm{Beta}(1,1)$; then the posterior mean $(s+1)/(n+2)$ is Laplace's rule of succession, and it equals the posterior predictive probability of a success, since $P(X_{\mathrm{new}} = 1 \mid D) = \int \theta\, p(\theta \mid D)\, d\theta$.

Example 2.9: the two sellers

Seller 1 has 90 positive and 10 negative reviews; seller 2 has 2 positive and 0 negative. With uniform priors

$$\theta_1 \mid D_1 \sim \mathrm{Beta}(91, 11), \qquad \theta_2 \mid D_2 \sim \mathrm{Beta}(3, 1).$$

Posterior predictive probabilities of a positive next review: $91/102 \approx 0.892$ versus $3/4 = 0.75$, so seller 1 is preferred, whereas the MLE (0.9 versus 1) prefers seller 2. A different question is which unknown rate is larger: since $\mathrm{Beta}(3,1)$ has CDF t^3,

$$P(\theta_1 > \theta_2 \mid D) = \mathbb{E}[\theta_1^3] = \frac{91 \cdot 92 \cdot 93}{102 \cdot 103 \cdot 104} \approx 0.713.$$

5 MAP as Regularisation and the Bias–Variance Decomposition

5.1 Gaussian mean with a Gaussian prior (Problem Set 1, Problem 2)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$ and prior $\theta \sim N(0, \tau^2)$.

Theorem 2.10: MAP is ridge, and it shrinks

$$\hat{\theta}_{\mathrm{MAP}} = \arg\min_{\theta} \sum_{i=1}^{n} (X_i - \theta)^2 + \lambda\theta^2, \quad \lambda = \frac{1}{\tau^2}, \qquad \hat{\theta}_{\mathrm{MAP}} = a\bar{X}, \quad a = \frac{n}{n+\lambda} = \frac{n\tau^2}{n\tau^2+1}.$$

Proof. $-\log p(\theta \mid X) = \frac{1}{2}\sum(X_i - \theta)^2 + \frac{\theta^2}{2\tau^2} + \text{const}$; multiplying by 2 gives the ridge objective. Setting the derivative $-2\sum(X_i - \theta) + 2\lambda\theta$ to zero gives $(n+\lambda)\theta = n\bar{X}$. ■

5.2 Bias–variance decomposition

Theorem 2.11: MSE = bias^2 + variance

For any estimator $\hat{\theta}$ of θ, $\mathbb{E}[(\hat{\theta} - \theta)^2] = (\mathbb{E}\hat{\theta} - \theta)^2 + \mathrm{Var}(\hat{\theta})$.

Proof. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}\hat{\theta}) + (\mathbb{E}\hat{\theta} - \theta)$. The second bracket is a constant and the first has mean zero, so the cross term vanishes after taking expectations. ■

For $\hat{\theta} = a\bar{X}$: bias $(a-1)\theta$, variance a^2/n, so $\mathrm{MSE}(a) = (a-1)^2\theta^2 + a^2/n$. Minimising over a gives $a^\star = n\theta^2/(n\theta^2+1)$, which equals the MAP shrinkage factor exactly when $\tau^2 = \theta^2$. Since $a^\star < 1$, some bias always pays off in MSE relative to the MLE $a = 1$; this is the first instance of the bias–variance trade-off that reappears for ridge regression (Chapter 4) and Adam's bias correction (Chapter 8).

Remark 2.12: Laplace prior gives the lasso

With $p(\theta) \propto e^{-|\theta|/b}$ instead, $-\log$ posterior is $\frac{1}{2}\sum(X_i-\theta)^2 + \frac{1}{b}|\theta|$ and the MAP is the soft threshold $\mathrm{sign}(\bar{X})\max(|\bar{X}| - \frac{1}{nb}, 0)$ (Chapter 4).

Chapter 3. Entropy, KL Divergence and Cross-Entropy

Entropy, KL divergence, mutual information, cross-entropy

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1 Information and Entropy

The information of an outcome is $h(x) = -\log p(x)$: rare outcomes carry more information, and for independent events probabilities multiply while information adds.

Definition 3.1: entropy

For a discrete random variable with pmf p,

$$H(X) = -\sum_x p(x)\log p(x) = \mathbb{E}[-\log p(X)], \qquad 0\log 0 := 0.$$

Natural logs give nats; base 2 gives bits.

Theorem 3.2: range of entropy

If X takes at most K values, $0 \le H(X) \le \log K$. $H = 0$ iff X is deterministic; $H = \log K$ iff X is uniform.

Proof. Each term $-p\log p \ge 0$ for $p \in [0,1]$, with equality iff $p \in \{0,1\}$. For the upper bound apply Jensen to the concave log: $H = \mathbb{E}\log\frac{1}{p(X)} \le \log \mathbb{E}\frac{1}{p(X)} = \log\sum_{x:p(x)>0} 1 \le \log K$, with equality iff $1/p(X)$ is constant on the support and the support has K points. ■

(For continuous densities the differential entropy can be negative; the statement above is for discrete variables.)

Definition 3.3: joint and conditional entropy

$$H(X,Y) = -\sum_{x,y} p(x,y)\log p(x,y),$$

$$H(Y\mid X) = -\sum_{x,y} p(x,y)\log p(y\mid x) = \sum_x p(x) H(Y\mid X=x).$$

Theorem 3.4: chain rule

$H(X,Y) = H(X) + H(Y\mid X)$.

Proof. Take $-\mathbb{E}$ of $\log p(X,Y) = \log p(X) + \log p(Y\mid X)$. ■

2 Kullback–Leibler Divergence

Definition 3.5: KL divergence

$$D_{\mathrm{KL}}(p\,\|\,q) = \sum_x p(x)\log\frac{p(x)}{q(x)} = \mathbb{E}_p\Big[\log\frac{p(X)}{q(X)}\Big].$$

Terms with $p(x) = 0$ contribute 0; if $p(x) > 0 = q(x)$ for some x the divergence is $+\infty$.

Theorem 3.6: Gibbs' inequality

$D_{\mathrm{KL}}(p\,\|\,q) \ge 0$, with equality iff $p = q$.

Proof. Sum over $S = \{x : p(x) > 0\}$. By Jensen for the concave log,

$$-D_{\mathrm{KL}}(p \,\|\, q) = \sum_{x \in S} p(x) \log \frac{q(x)}{p(x)} \le \log \sum_{x \in S} p(x) \frac{q(x)}{p(x)} = \log \sum_{x \in S} q(x) \le \log 1 = 0.$$

Equality forces (i) $q(x)/p(x) = c$ constant on S (Jensen) and (ii) $\sum_S q = 1$; then $c = \sum_S q = 1$, so $q = p$. Conversely $p = q$ gives $\log 1 = 0$ term by term. ■

Example 3.7: KL is not symmetric

For Bernoulli parameters, $D(r \,\|\, s) = r \log \frac{r}{s} + (1 - r) \log \frac{1-r}{1-s}$. With $r = 0.9$, $s = 0.5$: $D(r\|s) \approx 0.368$ but $D(s\|r) \approx 0.511$ nats. KL is not a metric (no symmetry, no triangle inequality).

3 Mutual Information

Definition 3.8: mutual information

$$I(X;Y) = D_{\mathrm{KL}}(p_{XY} \,\|\, p_X p_Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

Theorem 3.9: identities

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y) \ \ge 0,$$

with $I = 0$ iff $X \perp Y$. Consequently $H(X \mid Y) \le H(X)$ (conditioning reduces entropy).

Proof. $\log \frac{p(x,y)}{p(x)p(y)} = \log p(x \mid y) - \log p(x)$; take $\mathbb{E}$ and use $\sum_y p(x,y) = p(x)$ to get $-H(X \mid Y) + H(X)$. The last form follows from the chain rule. Nonnegativity and the equality case are Gibbs' inequality. ■

The entropy (Venn) diagram encodes $H(X,Y) = H(X \mid Y) + I(X;Y) + H(Y \mid X)$. For a fair bit X and $Y = X$, $I = 1$ bit; for independent fair bits, $I = 0$.

4 Cross-Entropy

Definition 3.10: cross-entropy

$H(p,q) = -\sum_x p(x) \log q(x) = \mathbb{E}_p[-\log q(X)]$.

Theorem 3.11: cross-entropy = entropy + KL

$H(p,q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)$. Hence for fixed p, $\arg\min_q H(p,q) = \arg\min_q D_{\mathrm{KL}}(p \,\|\, q) = p$.

Proof. $-\sum p \log q = -\sum p \log p + \sum p \log(p/q)$. The second statement uses Gibbs. ■

Remark 3.12: classification loss and soft labels

For a one-hot label $\mathbf{e}_y$, $H(\mathbf{e}_y) = 0$ and $H(\mathbf{e}_y, q) = D_{\mathrm{KL}}(\mathbf{e}_y \| q) = -\log q_y$: the classification loss. $H(p,q)$ is *linear in its first argument*, $H(\alpha p + \beta p', q) = \alpha H(p,q) + \beta H(p',q)$, because

p enters only as coefficients. Therefore a soft label that averages n one-hot labels gives $H(\frac{1}{n}\sum_j \mathbf{e}_{y_j}, q) = \frac{1}{n}\sum_j(-\log q_{y_j})$: fitting the label distribution by KL is equivalent (up to the constant $H(\hat{p})$) to ordinary cross-entropy on every individual label (2024 midterm Q2).

5 Jensen–Shannon Divergence (Problem Set 1, Problem 1)

Definition 3.13: JS divergence

With $m = \frac{1}{2}(p+q)$, $D_{\mathrm{JS}}(p \,\|\, q) = \frac{1}{2}D_{\mathrm{KL}}(p \,\|\, m) + \frac{1}{2}D_{\mathrm{KL}}(q \,\|\, m)$.

Theorem 3.14: properties

(i) $D_{\mathrm{JS}}(p\|q) \geq 0$, with equality iff $p = q$.

(ii) $D_{\mathrm{JS}}(p\|q) = D_{\mathrm{JS}}(q\|p)$.

(iii) $D_{\mathrm{JS}}(p\|q) = H(m) - \frac{1}{2}H(p) - \frac{1}{2}H(q)$.

(iv) $D_{\mathrm{JS}}(p\|q) \leq \log 2$, with equality iff p and q have disjoint supports.

Proof. (i) Both KL terms are nonnegative by Gibbs, so $D_{\mathrm{JS}} \geq 0$. If $D_{\mathrm{JS}} = 0$ then $D_{\mathrm{KL}}(p\|m) = 0$, so $p = m = \frac{1}{2}(p+q)$, i.e. $p = q$; conversely $p = q$ gives $m = p$ and both terms vanish. Note that $D_{\mathrm{KL}}(p\|m) < \infty$ always, since $m \geq p/2$. (ii) m is symmetric in p, q and the definition is a symmetric sum. (iii) $D_{\mathrm{KL}}(p\|m) = -H(p) - \sum p \log m$; averaging with the q term gives $-\frac{1}{2}H(p) - \frac{1}{2}H(q) - \sum m \log m$. (iv) Substituting m,

$$D_{\mathrm{JS}} = \log 2 + \tfrac{1}{2}\sum p \log \frac{p}{p+q} + \tfrac{1}{2}\sum q \log \frac{q}{p+q} \leq \log 2,$$

since $p/(p+q) \leq 1$ and $q/(p+q) \leq 1$. Equality iff $q = 0$ wherever $p > 0$ and vice versa. ■

Remark 3.15: JS as mutual information

Let B be a fair bit and $Z \sim p$ if $B = 0$, $Z \sim q$ if $B = 1$. Then $Z \sim m$, $H(Z \mid B) = \frac{1}{2}H(p) + \frac{1}{2}H(q)$, and by (iii) $I(Z;B) = D_{\mathrm{JS}}(p\|q) \leq H(B) = \log 2$.

Chapter 4. Probabilistic Regression, Regularisation and Kernel Ridge

Linear regression from a probabilistic view, regularisation, ridge regression, the kernel trick

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1 Gaussian Likelihood Explains Squared Error

Theorem 4.1: MLE under Gaussian noise is least squares

If $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, then

$$\log p(\mathbf{y} \mid X, \boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2.$$

For fixed σ^2, maximising the likelihood is minimising the SSE, so $\hat{\boldsymbol{\beta}}_{\text{MLE}}$ solves the normal equations. Maximising over σ^2 then gives $\hat{\sigma}^2 = \text{SSE}/n$.

Proof. The density of y_i is $(2\pi\sigma^2)^{-1/2}\exp\{-(y_i - \mathbf{x}_i^\top\boldsymbol{\beta})^2/(2\sigma^2)\}$; independence multiplies densities and the log turns the product into the displayed sum. Differentiating in σ^2: $-\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2\sigma^4} = 0$. ■

2 Ridge Regression

2.1 Definition and closed form

Definition 4.2: ridge

For $\lambda > 0$, $J(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$.

Theorem 4.3: ridge solution

$\nabla J = X^\top X\boldsymbol{\beta} - X^\top\mathbf{y} + \lambda\boldsymbol{\beta}$ and the unique minimiser is $\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda\mathbf{I})^{-1}X^\top\mathbf{y}$, even when X is rank deficient.

Proof. $\mathbf{v}^\top(X^\top X + \lambda\mathbf{I})\mathbf{v} = \|X\mathbf{v}\|^2 + \lambda\|\mathbf{v}\|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$, so the matrix is positive definite, invertible, and J is strictly convex. ■

2.2 MAP interpretation

With prior $\boldsymbol{\beta} \sim N(\mathbf{0}, \tau^2\mathbf{I})$, $-\log p(\boldsymbol{\beta} \mid \mathbf{y}) = \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{1}{2\tau^2}\|\boldsymbol{\beta}\|^2 + \text{const}$; multiplying by σ^2 gives ridge with $\lambda = \sigma^2/\tau^2$. A tight prior (small τ) means strong shrinkage.

2.3 Shrinkage in singular directions

Let $X = U\Sigma V^\top$ (thin SVD, singular values s_i). Since $X^\top X + \lambda\mathbf{I} = V(\Sigma^2 + \lambda\mathbf{I})V^\top$,

$$\hat{\boldsymbol{\beta}}_\lambda = \sum_i \frac{s_i}{s_i^2 + \lambda}(\mathbf{u}_i^\top\mathbf{y})\mathbf{v}_i, \qquad X\hat{\boldsymbol{\beta}}_\lambda = \sum_i \frac{s_i^2}{s_i^2 + \lambda}(\mathbf{u}_i^\top\mathbf{y})\mathbf{u}_i.$$

Least squares fits $\sum_i(\mathbf{u}_i^\top\mathbf{y})\mathbf{u}_i$; ridge multiplies the ith component by $s_i^2/(s_i^2 + \lambda) \in (0, 1)$. Small singular values (directions along which the data barely vary, where least squares amplifies noise by $1/s_i$) are shrunk most. $\lambda \to 0$ recovers least squares; $\lambda \to \infty$ sends coefficients to zero.

2.4 Bias and variance of ridge

Assume $\mathbf{y} = X\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$ with $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, and X of full column rank with $X^\top X = VDV^\top$, $D = \text{diag}(d_1, \ldots, d_p)$, $d_i > 0$.

Theorem 4.4: ridge trades bias for variance

(i) $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}_0 = -\lambda(X^\top X + \lambda \mathbf{I})^{-1}\boldsymbol{\beta}_0$; for $\lambda > 0$ the estimator is unbiased iff $\boldsymbol{\beta}_0 = \mathbf{0}$.

(ii) $\operatorname{Cov}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2(X^\top X + \lambda \mathbf{I})^{-1} X^\top X (X^\top X + \lambda \mathbf{I})^{-1}$ and

$$\operatorname{tr}\operatorname{Cov}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2 \sum_i \frac{d_i}{(d_i+\lambda)^2} < \sigma^2 \sum_i \frac{1}{d_i} = \operatorname{tr}\operatorname{Cov}(\hat{\boldsymbol{\beta}}_0) \qquad (\lambda > 0).$$

(iii) $\lambda \mapsto \|\hat{\boldsymbol{\beta}}_\lambda\|$ is decreasing.

Proof. (i) $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda \mathbf{I})^{-1} X^\top X \boldsymbol{\beta}_0$. From $(X^\top X + \lambda \mathbf{I})^{-1}(X^\top X + \lambda \mathbf{I}) = \mathbf{I}$ we get $(X^\top X + \lambda \mathbf{I})^{-1} X^\top X = \mathbf{I} - \lambda (X^\top X + \lambda \mathbf{I})^{-1}$, which gives the bias. The matrix $(X^\top X + \lambda \mathbf{I})^{-1}$ is invertible, so the bias vanishes iff $\boldsymbol{\beta}_0 = \mathbf{0}$. (ii) $\hat{\boldsymbol{\beta}}_\lambda = M\mathbf{y}$ with $M = (X^\top X + \lambda \mathbf{I})^{-1} X^\top$, so $\operatorname{Cov} = M(\sigma^2 \mathbf{I}) M^\top$. All three matrices are diagonalised by V, giving eigenvalues $\sigma^2 d_i/(d_i+\lambda)^2$; the trace is their sum, and $(d_i+\lambda)^2 > d_i^2$. (iii) $\|\hat{\boldsymbol{\beta}}_\lambda\|^2 = \sum_i s_i^2 (\mathbf{u}_i^\top \mathbf{y})^2/(s_i^2+\lambda)^2$ by orthonormality of $\mathbf{v}_i$; each term decreases in λ. ■

3 Other Penalties and the Lasso

More generally one penalises $\sum_j |\beta_j|^q$. $q = 2$ is ridge (round level sets, pure shrinkage); $q = 1$ is the lasso, whose diamond-shaped level sets have corners on the axes and produce exact zeros; $0 < q < 1$ is nonconvex.

Theorem 4.5: soft thresholding

$\arg\min_b \ \frac{1}{2}(b-z)^2 + \lambda|b| = \operatorname{sign}(z)\max(|z|-\lambda, 0)$, whereas $\arg\min_b \frac{1}{2}(b-z)^2 + \frac{\lambda}{2}b^2 = z/(1+\lambda)$.

Proof. The objective is strictly convex; the optimality condition is $0 \in b - z + \lambda\, \partial|b|$, where $\partial|b| = \{\operatorname{sign} b\}$ for $b \neq 0$ and $[-1, 1]$ at 0. If $b > 0$: $b = z - \lambda$, valid iff $z > \lambda$. If $b < 0$: $b = z + \lambda$, valid iff $z < -\lambda$. If $b = 0$: need $z \in \lambda[-1, 1]$, i.e. $|z| \leq \lambda$. The ridge case is the stationarity condition $b - z + \lambda b = 0$. ■

Lasso moves every coefficient by a fixed amount and cuts small ones to zero (sparsity); ridge rescales proportionally and never produces exact zeros.

4 Kernel Ridge Regression

4.1 Feature-space and sample-space forms

Let $\boldsymbol{\phi}(\mathbf{x}) \in \mathbb{R}^m$ be a feature map and, $\Phi = [\boldsymbol{\phi}(\mathbf{x}_1) \cdots \boldsymbol{\phi}(\mathbf{x}_n)] \in \mathbb{R}^{m \times n}$ (columns are samples). Ridge in feature space minimises $J = \frac{1}{2}\|\mathbf{y} - \Phi^\top \mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$, so

$$\nabla J = -\Phi\mathbf{y} + (\lambda \mathbf{I}_m + \Phi\Phi^\top)\mathbf{w}, \qquad \hat{\mathbf{w}} = (\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi\mathbf{y}.$$

Theorem 4.6: push-through identity

$(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi = \Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$. Hence with the Gram matrix $K = \Phi^\top\Phi$, $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j) = \boldsymbol{\phi}(\mathbf{x}_i)^\top \boldsymbol{\phi}(\mathbf{x}_j)$,

$$\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha}, \quad \boldsymbol{\alpha} = (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}, \qquad f(\mathbf{x}) = \boldsymbol{\phi}(\mathbf{x})^\top \hat{\mathbf{w}} = \sum_{i=1}^n \alpha_i k(\mathbf{x}_i, \mathbf{x}) = \mathbf{k}_\mathbf{x}^\top (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}.$$

Proof. $\Phi(\lambda \mathbf{I}_n + \Phi^\top \Phi) = \lambda\Phi + \Phi\Phi^\top\Phi = (\lambda \mathbf{I}_m + \Phi\Phi^\top)\Phi$. Multiply on the left by $(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}$ and on the right by $(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$ (both positive definite). The two identity matrices have different sizes. ■

Remark 4.7: the kernel trick and cost

$\boldsymbol{\phi}$ appears only through inner products, so only the kernel k is needed. The primal form solves an $m \times m$ system and the dual an $n \times n$ one: the kernel form wins when $m \gg n$ (e.g. the Gaussian kernel, whose feature space is infinite-dimensional). Merely mapping to higher dimensions is feature engineering; computing through k without forming $\boldsymbol{\phi}$ is the kernel trick. An alternative derivation substitutes $\mathbf{w} = \Phi\boldsymbol{\alpha}$ and obtains $K[(K + \lambda \mathbf{I})\boldsymbol{\alpha} - \mathbf{y}] = \mathbf{0}$; when K is singular one may not cancel K, but the displayed $\boldsymbol{\alpha}$ is a solution and all solutions give the same $\hat{\mathbf{w}}$.

4.2 Kernels

- Linear: $k(\mathbf{x}, \mathbf{z}) = \mathbf{x}^\top \mathbf{z}$.
- Quadratic in 2D: $(\mathbf{x}^\top \mathbf{z})^2 = x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 x_2 z_1 z_2$, so $\boldsymbol{\phi}(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1 x_2)$. In 1D, $(1 + xz)^2$ has $\boldsymbol{\phi}(x) = (1, \sqrt{2}x, x^2)$.
- Gaussian: $k(\mathbf{x}, \mathbf{z}) = \exp(-\|\mathbf{x} - \mathbf{z}\|^2/(2\sigma^2))$; σ is the bandwidth.
- Normalised: $\tilde{k}(\mathbf{x}, \mathbf{z}) = k(\mathbf{x}, \mathbf{z})/\sqrt{k(\mathbf{x}, \mathbf{x})k(\mathbf{z}, \mathbf{z})}$.

Theorem 4.8: Gram matrices are PSD

For any feature-map kernel and any $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a}^\top K \mathbf{a} = \|\sum_i a_i \boldsymbol{\phi}(\mathbf{x}_i)\|^2 \geq 0$.

Not every similarity function is a valid kernel; positive semidefiniteness of all Gram matrices is the requirement.

4.3 Worked examples

Example 4.9: linear kernel

$\mathbf{x}_1 = (1, 2)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 1)$, $\lambda = 1$: $K = \begin{pmatrix} 5 & 2 \\ 2 & 4 \end{pmatrix}$, $(K + \mathbf{I})^{-1} = \frac{1}{26}\begin{pmatrix} 5 & -2 \\ -2 & 6 \end{pmatrix}$, $\boldsymbol{\alpha} = (1/2, 0)$, $\hat{\mathbf{w}} = \frac{1}{2}\mathbf{x}_1 = (1/2, 1)$. At $\mathbf{x} = (1, 0)$: $\mathbf{k}_\mathbf{x} = (1, 2)$ and $f(\mathbf{x}) = 1/2 = \hat{\mathbf{w}}^\top \mathbf{x}$.

Example 4.10: quadratic kernel

$\mathbf{x}_1 = (1, 1)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 2)$, $k = (\mathbf{x}^\top \mathbf{z})^2$, $\lambda = 1$. Then $K_{11} = 4$, $K_{12} = 4$, $K_{22} = 16$,

$$(K + \mathbf{I})^{-1} = \frac{1}{69}\begin{pmatrix} 17 & -4 \\ -4 & 5 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \frac{1}{69}\begin{pmatrix} 43 \\ -2 \end{pmatrix}.$$

At $\mathbf{x}_t = (1, 0)$, $\mathbf{k}_{\mathbf{x}_t} = (1, 4)$ and $f(\mathbf{x}_t) = (43 - 8)/69 = 35/69 \approx 0.507$. Check in feature space: $\boldsymbol{\phi}(\mathbf{x}_1) = (1, 1, \sqrt{2})$, $\boldsymbol{\phi}(\mathbf{x}_2) = (4, 0, 0)$, $\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha} = (35, 43, 43\sqrt{2})/69$ and $\boldsymbol{\phi}(\mathbf{x}_t) = (1, 0, 0)$ give the same $35/69$.

Chapter 5. Classification: Logistic Regression, Softmax and SVM

Logistic regression, multiclass classification, support vector machines

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1 Odds, Logit and the Logistic Model

For a probability p, the odds are $o = p/(1-p)$ ($p = 0.8$ gives odds 4) and

$$\operatorname{logit}(p) = \log \frac{p}{1-p} \in \mathbb{R}, \qquad \sigma(z) = \frac{1}{1+e^{-z}} = \operatorname{logit}^{-1}(z).$$

Definition 5.1: logistic regression

With augmented input $\mathbf{x} = (1, x_1, \ldots, x_d)$ and $y \in \{0, 1\}$,

$$\log \frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}, \qquad P(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}).$$

The score is linear and the probability transform nonlinear. Thresholding the probability at τ gives the hyperplane $\mathbf{w}^\top \mathbf{x} = \log \frac{\tau}{1-\tau}$ (at $\tau = 0.5$: $\mathbf{w}^\top \mathbf{x} = 0$).

Theorem 5.2: sigmoid identities

$\sigma'(z) = \sigma(z)(1-\sigma(z)) \le \frac{1}{4}$, $1 - \sigma(z) = \sigma(-z)$, $\frac{d}{dz} \log(1+e^z) = \sigma(z)$.

Proof. $\sigma'(z) = e^{-z}/(1+e^{-z})^2 = \sigma(z) \cdot \frac{e^{-z}}{1+e^{-z}}$ and $\frac{e^{-z}}{1+e^{-z}} = \frac{1}{e^z+1} = \sigma(-z) = 1 - \sigma(z)$. The maximum of $s(1-s)$ on $[0,1]$ is $1/4$ at $s = 1/2$. ■

2 Likelihood, Gradient, Hessian and Convexity

For data $(\mathbf{x}_i, y_i)$ with $p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = p_i^{y_i}(1-p_i)^{1-y_i}$ and

$$J(\mathbf{w}) = -\sum_i \left[y_i \log p_i + (1-y_i) \log(1-p_i) \right] = \sum_i \left[\log(1+e^{z_i}) - y_i z_i \right], \quad z_i = \mathbf{w}^\top \mathbf{x}_i.$$

Theorem 5.3: logistic loss is convex (2025 midterm Q1)

$\nabla J = \sum_i (p_i - y_i)\mathbf{x}_i = X^\top(\mathbf{p} - \mathbf{y})$ and $\nabla^2 J = \sum_i p_i(1-p_i)\mathbf{x}_i\mathbf{x}_i^\top = X^\top S X$, $S = \operatorname{diag}(p_i(1-p_i))$. Hence J is convex, and strictly convex if X has full column rank.

Proof. $\partial J/\partial z_i = \sigma(z_i) - y_i = p_i - y_i$ by the identities above, and $\partial z_i/\partial \mathbf{w} = \mathbf{x}_i$. Differentiating again, $\nabla p_i = p_i(1-p_i)\mathbf{x}_i$. For any $\mathbf{v}$, $\mathbf{v}^\top X^\top S X \mathbf{v} = \sum_i p_i(1-p_i)(\mathbf{x}_i^\top \mathbf{v})^2 \ge 0$; since $p_i(1-p_i) > 0$, the sum is positive whenever $X\mathbf{v} \neq \mathbf{0}$. ■

Example 5.4: one gradient step

$\mathbf{x} = (1, 2)$, $y = 1$, $\mathbf{w} = \mathbf{0}$: $p = 1/2$, $\nabla J = (p-1)\mathbf{x} = (-1/2, -1)$. With $\eta = 0.1$, $\mathbf{w} = (0.05, 0.1)$, score 0.25, $p \approx 0.562$: the probability of the correct label increased.

3 Labels in $\{\pm 1\}$, Logistic Loss and BCE

With $Y \in \{\pm 1\}$ and score $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$, $P(Y = +1 \mid \mathbf{x}) = \sigma(f)$ and $P(Y = -1 \mid \mathbf{x}) = \sigma(-f)$, so $-\log P(Y \mid \mathbf{x}) = \ell(Yf)$ with the **logistic loss** $\ell(z) = \log(1+e^{-z})$. $Yf(\mathbf{x})$ is the *margin*: positive iff the sign of the score is correct.

Theorem 5.5: three names for one objective (2022 midterm Q4)

Let $P(-1) = (1,0)$, $P(+1) = (0,1)$, $\mu(r) = (\sigma(-r), \sigma(r))$, $\tilde{Y} = (1+Y)/2 \in \{0,1\}$ and $\mathrm{BCE}(s,t) = -[t \log s + (1-t)\log(1-s)]$. For every $(\mathbf{a}, b)$ and every i,

$$D_{\mathrm{KL}}(P(Y_i) \,\|\, \mu(f(\mathbf{x}_i))) = \mathrm{BCE}\left(\sigma(f(\mathbf{x}_i)), \tilde{Y}_i\right) = \ell(Y_i f(\mathbf{x}_i)).$$

So KL minimisation, BCE minimisation and logistic regression are the same problem.

Proof. For a one-hot first argument, KL $= -\log($model probability of the true class$)$. If $Y = +1$ this is $-\log\sigma(r) = \log(1+e^{-r}) = \ell(r)$, and BCE with $t = 1$ is $-\log s$. If $Y = -1$ it is $-\log\sigma(-r) = \ell(-r)$, and BCE with $t = 0$ is $-\log(1-s) = -\log\sigma(-r)$. ■

Theorem 5.6: ℓ is convex

$\ell'(z) = \sigma(z) - 1$ and $\ell''(z) = \sigma(z)\sigma(-z) > 0$.

3.1 Separable data (2024 midterm Q3)

Theorem 5.7: no MLE on separable data

Suppose $Y_i(\mathbf{a}^{\star\top} X_i + b^\star) > 0$ for all i. Then $\inf_{\mathbf{a},b} \sum_i \ell(Y_i(\mathbf{a}^\top X_i + b)) = 0$, the infimum is not attained, and every minimising sequence is unbounded. With an ℓ_2 penalty $\frac{\lambda}{2}(\|\mathbf{a}\|^2 + b^2)$ a unique minimiser exists and satisfies $\|\mathbf{a}_\lambda\|^2 + b_\lambda^2 \le 2N\log 2/\lambda$.

Proof. Let $t = \min_i Y_i(\mathbf{a}^{\star\top} X_i + b^\star) > 0$ (finitely many positive numbers). Scaling by $C > 0$ and using that ℓ is decreasing, $L(C\mathbf{a}^\star, Cb^\star) \le N\log(1 + e^{-Ct}) \to 0$; and $L \ge 0$. Every term is strictly positive, so $L > 0$ everywhere. If a minimising sequence were bounded, a convergent subsequence would give a point with $L = 0$. With the penalty: $L_\lambda \ge \frac{\lambda}{2}\|(\mathbf{a}, b)\|^2$ makes the sublevel set $\{L_\lambda \le L_\lambda(\mathbf{0})\}$ compact, so a minimiser exists (Weierstrass); strong convexity gives uniqueness; and $\frac{\lambda}{2}\|(\mathbf{a}_\lambda, b_\lambda)\|^2 \le L_\lambda(\mathbf{0}) = N\ell(0) = N\log 2$. ■

3.2 Surrogate risk bounds (2025 midterm Q5)

For $(X, Y) \sim P$ and classifier $f = \mathrm{sign}(g)$, counting $g = 0$ as an error, the event $\{f(X) \ne Y\}$ implies $Yg(X) \le 0$. For a nonnegative, nonincreasing loss ϕ this gives $\phi(Yg) \ge \phi(0)$ on errors and therefore

$$P(f(X) \ne Y) \le \frac{\mathbb{E}[\phi(Yg(X))]}{\phi(0)}.$$

Logistic: $\phi(0) = \log 2$, so $R_{\log} < \varepsilon \Rightarrow P(f(X) = Y) > 1 - \varepsilon/\log 2$. Hinge $\max(0, 1-u)$ and exponential e^{-u} have $\phi(0) = 1$. Minimising a convex surrogate thus controls the 0–1 error, which itself cannot be optimised by gradients.

4 Multiclass Classification and Softmax

For Q classes, $W \in \mathbb{R}^{Q\times d}$, scores $s_k = \mathbf{w}_k^\top \mathbf{x}$ and

$$p_k(\mathbf{x}) = \mathrm{softmax}(\mathbf{s})_k = \frac{e^{s_k}}{\sum_r e^{s_r}}, \qquad J(W) = -\sum_{i,k} y_{ik} \log p_k(\mathbf{x}_i).$$

Scores $(2, 1, 0)$ give approximately $(0.665, 0.245, 0.090)$.

Theorem 5.8: softmax calculus

With $\mathrm{LSE}(\mathbf{s}) = \log \sum_r e^{s_r}$ and $\mathbf{p} = \mathrm{softmax}(\mathbf{s})$:

(i) $\mathrm{softmax}(\mathbf{s} + c\mathbf{1}) = \mathrm{softmax}(\mathbf{s})$, $\mathrm{LSE}(\mathbf{s} + c\mathbf{1}) = \mathrm{LSE}(\mathbf{s}) + c$; stable evaluation $\mathrm{LSE}(\mathbf{s}) = M + \log \sum e^{s_r - M}$, $M = \max s_r$.

(ii) $\nabla \mathrm{LSE} = \mathbf{p}$; Jacobian $\partial p_k / \partial s_m = p_k(\delta_{km} - p_m)$, i.e. $\nabla^2 \mathrm{LSE} = \mathrm{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top \succeq 0$.

(iii) $\nabla_{\mathbf{s}}(-\log p_y) = \mathbf{p} - \mathbf{e}_y$; for the whole data set $\nabla_W J = (P - Y)^\top X$ (rows: classes).

(iv) $\log \mathrm{softmax}(\mathbf{s}) = \mathbf{s} - \mathrm{LSE}(\mathbf{s})\mathbf{1}$ and log-softmax is idempotent: $\log \mathrm{softmax}(\log \mathrm{softmax}(\mathbf{s})) = \log \mathrm{softmax}(\mathbf{s})$ (2024 midterm Q4).

(v) $\max_k s_k \le \mathrm{LSE}(\mathbf{s}) \le \max_k s_k + \log Q$.

Proof. (i) e^c cancels. (ii) Quotient rule; $\mathbf{v}^\top(\mathrm{diag}\,\mathbf{p} - \mathbf{p}\mathbf{p}^\top)\mathbf{v} = \sum p_k v_k^2 - (\sum p_k v_k)^2$ is the variance of v_K with $K \sim \mathbf{p}$. (iii) $-\log p_y = -s_y + \mathrm{LSE}(\mathbf{s})$. (iv) With $\boldsymbol{\ell} = \log \mathrm{softmax}(\mathbf{s})$, $e^{\ell_j} = p_j$ so $\mathrm{LSE}(\boldsymbol{\ell}) = \log \sum p_j = 0$. (v) $e^M \le \sum e^{s_r} \le Q e^M$. ■

Adding the same vector to all rows of W leaves the softmax unchanged, so W is identifiable only up to such shifts.

5 Support Vector Machines

5.1 Geometry and the primal problem

Labels are $y_i \in \{\pm 1\}$, score $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$. The distance from $\mathbf{x}_0$ to $\{\mathbf{w}^\top \mathbf{x} + b = 0\}$ is $|\mathbf{w}^\top \mathbf{x}_0 + b|/\|\mathbf{w}\|$ (move along the unit normal: $\mathbf{x}_0 - t\mathbf{w}/\|\mathbf{w}\|$ lies on the plane iff $t = (\mathbf{w}^\top \mathbf{x}_0 + b)/\|\mathbf{w}\|$). Rescaling $(\mathbf{w}, b)$ by $c > 0$ changes nothing, so for separable data normalise the closest functional margin to 1:

Definition 5.9: hard-margin SVM

$$\min_{\mathbf{w},b} \ \tfrac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \ge 1, \ i = 1, \dots, n.$$

The support planes $\mathbf{w}^\top \mathbf{x} + b = \pm 1$ lie at distance $1/\|\mathbf{w}\|$ (the margin) from the boundary; the slab width is $2/\|\mathbf{w}\|$.

5.2 Lagrangian, max–min inequality, weak duality

Theorem 5.10: max–min inequality (Problem Set 1, Problem 4)

For any $F : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, $\sup_x \inf_y F(x, y) \le \inf_y \sup_x F(x, y)$. Equality can fail: for $F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (rows x, columns y) the left side is 0 and the right side 1.

Proof. For all x, y: $\inf_{y'} F(x, y') \le F(x, y) \le \sup_{x'} F(x', y)$. The left end does not depend on y and the right end not on x; take $\sup_x$ on the left, then $\inf_y$ on the right. ■

With $c_i = 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)$ and $\alpha_i \ge 0$, $L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_i \alpha_i c_i$. Since $\sup_{\alpha \ge 0} \alpha c = 0$ if $c \le 0$ and $+\infty$ if $c > 0$, $\sup_{\boldsymbol{\alpha} \ge \mathbf{0}} L$ equals the objective on feasible points and $+\infty$ elsewhere; hence $p^\star = \inf_{\mathbf{w},b} \sup_{\boldsymbol{\alpha}} L$. The dual is $d^\star = \sup_{\boldsymbol{\alpha}} \inf_{\mathbf{w},b} L$ and the max–min inequality gives **weak duality**

$d^\star \le p^\star$: every dual value is a lower bound on the primal optimum. For the SVM (convex, strictly feasible) strong duality $d^\star = p^\star$ holds.

5.3 The dual problem

Theorem 5.11: SVM dual

$\nabla_{\mathbf{w}} L = \mathbf{0} \Rightarrow \mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$; finiteness of $\inf_b L$ requires $\sum_i \alpha_i y_i = 0$. The dual is

$$\max_{\boldsymbol{\alpha} \ge \mathbf{0}} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0.$$

Complementary slackness $\alpha_i [1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)] = 0$: $\alpha_i > 0$ only for support vectors, and $b = y_i - \mathbf{w}^\top \mathbf{x}_i$ for any such i.

Proof. b enters L linearly with coefficient $-\sum \alpha_i y_i$; if nonzero, $\inf_b = -\infty$. Substituting $\mathbf{w}$: $\frac{1}{2}\|\mathbf{w}\|^2 - \|\mathbf{w}\|^2 + \sum \alpha_i = \sum \alpha_i - \frac{1}{2}\|\mathbf{w}\|^2$. ■

Example 5.12: two points (Problem Set 1, Problem 3)

$\mathbf{x}_1 = (0,0)$, $y_1 = -1$; $\mathbf{x}_2 = (2,2)$, $y_2 = +1$. The constraint forces $\alpha_1 = \alpha_2 = a$ and $\|\mathbf{x}_2\|^2 = 8$, so the dual is $2a - 4a^2$, maximised at $a = 1/4$. Then $\mathbf{w} = \frac{1}{4}(2,2) = (1/2, 1/2)$, $b = -1$ (from $\mathbf{x}_1$), the boundary is $x^{(1)} + x^{(2)} = 2$, $\|\mathbf{w}\| = 1/\sqrt{2}$, margin $\sqrt{2}$ (slab width $2\sqrt{2}$). Primal value $\frac{1}{2}\|\mathbf{w}\|^2 = 1/4$ equals the dual value $2a - 4a^2 = 1/4$.

Remark 5.13: soft margin and kernels

For non-separable data add slacks: $\min \frac{1}{2}\|\mathbf{w}\|^2 + C\sum \xi_i$ s.t. $y_i f(\mathbf{x}_i) \ge 1 - \xi_i$, $\xi_i \ge 0$, equivalently hinge loss $C\sum \max(0, 1 - y_i f(\mathbf{x}_i))$ plus ℓ_2. Its dual is identical except for the box constraint $0 \le \alpha_i \le C$. Replacing $\mathbf{x}_i^\top \mathbf{x}_j$ by a kernel gives the kernel SVM with score $\sum_i \alpha_i y_i k(\mathbf{x}_i, \mathbf{x}) + b$. Logistic regression fits a likelihood; the SVM maximises a geometric margin.